EQUALITY CRITERION FOR MORPHISMS IN THE LOCALIZED CATEGORY WITH THE SAME DENOMINATOR

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 4536-4577, following the statement at 4520-4534.

Editorial note. The fraction rules used in the chain computation are those of the definition discussion referenced at 4569-4570.

Allowed standard prerequisites (editorial).

- Localization category is well defined. Lemma lemma-left-localization of this chapter, invoked at proof lines 4537-4538 (carded as L26112). S^{-1}C is a category, so compositions and equality are meaningful.
- Fraction notation and composition rules. Definition definition-left-localization-as-fraction of this chapter and the discussion following it, used at 4541 and 4569-4576. Writes A = s^{-1}f and computes id^{-1}_{Y"} s' o s^{-1} f = id^{-1}_{Y"}(a o f).

Source reference. Original definition 0BM2: sources/stacks-categories/pinned-original-source.tex, source lines 4456–4456. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 04VD: sources/stacks-categories/pinned-original-source.tex, source lines 4219–4219. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

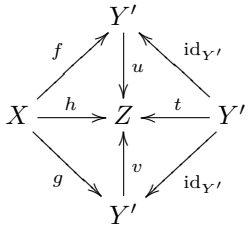
Lemma 0.1. *Let $\mathcal{C}$ be a category and let S be a left multiplicative system of morphisms of $\mathcal{C}$. Let $A, B : X \to Y$ be morphisms of $S^{-1}\mathcal{C}$ which are the equivalence classes of $(f : X \to Y', s : Y \to Y')$ and $(g : X \to Y', s : Y \to Y')$. The following are equivalent*

(1) *$A = B$*
(2) *there exists a morphism $t : Y' \to Y''$ in S with $t \circ f = t \circ g$, and*
(3) *there exists a morphism $a : Y' \to Y''$ such that $a \circ f = a \circ g$ and $a \circ s \in S$.*

Proof. We are going to use that $S^{-1}\mathcal{C}$ is a category (Lemma 04VD) and we will use the notation of Definition 0BM2 as well as the discussion following that definition to identify some morphisms in $S^{-1}\mathcal{C}$. Thus we write $A = s^{-1}f$ and $B = s^{-1}g$.

Date: stacks-project:categories.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

If $A = B$ then $(\mathrm{id}_{Y'}^{-1}s) \circ A = (\mathrm{id}_{Y'}^{-1}s) \circ B$. We have $(\mathrm{id}_{Y'}^{-1}s) \circ A = \mathrm{id}_{Y'}^{-1}f$ and $(\mathrm{id}_{Y'}^{-1}s) \circ B = \mathrm{id}_{Y'}^{-1}g$. The equality of $\mathrm{id}_{Y'}^{-1}f$ and $\mathrm{id}_{Y'}^{-1}g$ means by definition that there exists a commutative diagram



with $t \in S$. In particular $u = v = t \in S$ and $t \circ f = t \circ g$. Thus (1) implies (2).

The implication (2) $\Rightarrow$ (3) is immediate. Assume a is as in (3). Denote $s' = a \circ s \in S$. Then $\mathrm{id}_{Y''}^{-1}s'$ is an isomorphism in the category $S^{-1}\mathcal{C}$ (with inverse $(s')^{-1}\mathrm{id}_{Y''}$). Thus to check $A = B$ it suffices to check that $\mathrm{id}_{Y''}^{-1}s' \circ A = \mathrm{id}_{Y''}^{-1}s' \circ B$. We compute using the rules discussed in the text following Definition 0BM2 that $\mathrm{id}_{Y''}^{-1}s' \circ A = \mathrm{id}_{Y''}^{-1}(a \circ s) \circ s^{-1}f = \mathrm{id}_{Y''}^{-1}(a \circ f) = \mathrm{id}_{Y''}^{-1}(a \circ g) = \mathrm{id}_{Y''}^{-1}(a \circ s) \circ s^{-1}g = \mathrm{id}_{Y''}^{-1}s' \circ B$ and we see that (1) is true. $\square$